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Testování efektu AI Foundational models na analýzu NGS dat DNA sekvenování

Diplomová práce

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Vedoucí práce: Mgr. Vojtěch Bystrý, Ph.D. Brno 2026

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Abstrakt

V této bakalářské/diplomové/rigorózní práci se věnujeme ...

Abstract

In this thesis we study ...

Místo tohoto listu vložte kopii oficiálního zadání práce bez podpisů.

Poděkování

Na tomto místě bych chtěl(-a) poděkovat ...

Prohlášení

Prohlašuji, že jsem svouji diplomovou práci vypracoval samostatně pod vedením vedoucího práce s využitím informačních zdrojů, které jsou v práci citovány. Prohlašuji, že jsem pro stylistické úpravy textu využil nástroje AI a to plně v souladu s principy akademické integrity.

Brno 13. Května 2026

. . . .
Tomáš Vamberský

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List of abbreviations

$\mathbb{C}$ množina všech komplexních čísel
$\mathbb{R}$ množina všech reálných čísel
$\mathbb{Z}$ množina všech celých čísel
$\mathbb{N}$ množina všech přirozených čísel

Introduction

Chapter 1

Theoretical background

1.1 Transcription factors

1.1.1 Transcription factors and DNA binding

Transcription factors (TFs) are proteins that modulate transcription by binding DNA in a sequence-specific manner. The target sequences are typically motifs of 6–20 bp and are recognized with tolerance for sequence variation. They act through regulatory elements such as promoters, enhancers, and silencers, by recruiting the transcription machinery or modulating its activity. Most TFs share a modular architecture: a DNA-binding domain (DBD) that contacts the target sequence, paired with one or more effector domains responsible for transactivation, repression, or dimerisation. Because the DBD determines binding specificity, TFs are usually classified by DBD family rather than by their effector domains or biological function. The human genome encodes approximately 1600 sequence-specific TFs, which can be classified into roughly 75 DBD families (Lambert et al., 2018). These families include the C2H2 zinc fingers (the family most represented in humans, e.g. CTCF), the basic helix-loop-helix proteins (e.g. MYC, which binds as a heterodimer with MAX), the ETS domain family (e.g. SPI1), and the forkhead family (e.g. FOXK2).

Despite the considerable variety in structure, all sequence-specific TFs face the same problem - they must recognise their short target sequence among numerous similar but non-functional sites in a genome.

1.1.2 Sequence determinants of binding

The specificity of protein-DNA binding is based on two complementary mechanisms: direct chemical recognition of bases (base readout), and recognition of sequence-dependent DNA geometry (shape readout).

Base readout works through direct contact, either through hydrogen bonds or van der Waals interactions, of amino acid side chains and the Watson-Crick edges of bases, predominantly in the major groove. The major groove presents a distinctive pattern of hydrogen bond donors and acceptors that allows discrimination between base pairs, including for example the orientation of A-T or T-A (Luscombe et al., 2001; Seeman et al., 1976). The canonical example of base readout is the C2H2 zinc finger, in which each $\beta\beta\alpha$ module contacts approximately three base pairs via two or three specificity residues on the recognition helix, contributing additively to the binding affinity of the multi-finger array. Position weight matrices (PWMs) are the natural statistical encoding of this mechanism, treating each position as an independent contribution to binding affinity.

Unlike base readout, shape readout depends on sequence context beyond the contact position - the geometry at any given base pair is influenced by neighbouring bases over several positions in either direction. The relevant geometric features (minor groove width, propeller twist, helical twist, roll, and slide...) describe the orientation and stacking of base pairs along the helix and are not direct chemical signals of base identity. They are read by protein contacts that respond to local shape and electrostatics rather than to specific bases. The canonical example is recognition of narrow minor grooves by arginine residues: A-tracts (runs of A-T base pairs) produce locally narrowed minor grooves with concentrated negative electrostatic potential, which arginines insert into and stabilise without forming base-specific hydrogen bonds (Rohs et al., 2009).

Even taken together, base and shape readout cannot fully account for the specificity of in vivo TF binding. Both mechanisms are short-range: base readout depends on direct contacts within the binding footprint of typically 6–20 bp, and shape readout, while sensitive to surrounding sequence, is mediated through contacts within and immediately flanking that footprint (Yang et al., 2017). Yet these mechanisms are genuinely complementary rather than redundant. Crystal structures of TF–DNA complexes routinely show major-groove base-specific contacts and minor-groove shape-based contacts within the same interface, and quantitative analyses show that augmenting PWM-based models with shape features systematically improves binding prediction across TF families (Mathelier et al., 2016). *The limitation is not that the two mechanisms fail individually, but that their combined reach is bounded by the binding footprint. The discrepancy between motif occurrences in the genome and the subset that is actually bound — addressed in the next paragraph — therefore demands explanations beyond the recognition mechanisms reviewed here.*

Sequence-specific TFs recognise short DNA motifs, typically 6–20 bp long. The recognition is degenerate: most positions tolerate multiple bases, with information formally captured by position weight matrices [TODO: odkaz na dalsi kapitolu? cut protoze se opakuji?]. The motif as a whole specifies the binding sequence much less precisely than its nominal length would suggest. Sequence recognition mechanisms reviewed above operate over tens of base pairs at most. In the context of the 3 Gb human genome, hundreds of thousands to millions of matches are expected by chance alone. ChIP-seq, the standard assay for in vivo TF occupancy (described in the next section), typically identifies between a few thousand and a few tens of thousands of bound sites per TF per cell line. CTCF in K562 cells, for example, has approximately 52,000 IDR-thresholded ENCODE peaks, against approximately 870,000 matches to the canonical CTCF motif in the hg19 genome at standard FIMO thresholds (stricter thresholds reduce this to a few tens of thousands, but at the cost of missing many genuine binding sites) (Dozmorov et al., 2022). This gap between motif presence and actual binding has been termed the transcription factor specificity paradox, *originally formalised for prokaryotic TFs as* *the futility theorem by Wunderlich a Mirny (2009): in a sufficiently large genome, even highly informative motifs predict more matches than there are TF molecules in the cell, making purely sequence-based recognition statistically insufficient.* In practice, this is reflected in the weak correlation between intrinsic in vitro affinity (measured by HT-SELEX or protein-binding microarrays) and in vivo occupancy (measured by ChIP-seq): high-affinity sites can remain unbound, while low-affinity sites are often bound (Slattery et al., 2014). This gap is partially explained by several mechanisms, each a substantial research field in its own right.

Chromatin accessibility is regarded as the dominant factor: only a small fraction of the genome may be accessed by proteins at any given time, the rest made unavailable by nucleosomes or compacted into heterochromatin. ATAC-seq, DNase-seq, and FAIRE-seq are used to identify accessible , and most in vivo TF binding occurs within it (Thurman et al., 2012). *Cooperative binding with cofactors is the second. Many TFs bind productively only as part of a complex — homodimers, heterodimers, or larger assemblies — so combinatorics restrict where binding can occur (Spitz a Furlong, 2012)[nemam access, najit lepsi].* MYC, for example, binds as an obligate heterodimer with MAX. Cellular context further narrows the set: TF abundance, post-translational modification, and subcellular localisation are all regulated and condition-dependent.

The fourth and final mechanism - sequence context beyond the motif - is the focus of this thesis. Flanking nucleotides, dinucleotide composition, DNA shape, and broader compositional features of the regions surrounding a motif modulate binding affinity and contribute to the discrimination between bound and unbound matches (Dror et al., 2015). Unlike chromatin and cooperativity, this contribution is encoded directly in the DNA sequence and is therefore in principle accessible to any model that operates on raw sequence input. The next paragraph develops what is currently known about this contribution, from the immediate flanks of the motif out to the kilobase-scale compositional gradients recently characterised by Faltejsková et al. (2026) — the biological signal that motivates the rest of this thesis.

The fact that motif occurrences vastly outnumber bound sites motivates a substantial body of work on what else in the DNA sequence influences in vivo binding. The contributions that have been characterised span three orders of magnitude in distance.

At the shortest scale, the nucleotides immediately flanking the core motif modify binding affinity in ways that PWMs cannot capture. (Dror et al., 2015) showed that flanking-region features improve TF binding prediction across diverse families, with much of the gain attributable to DNA shape: the flanks determine the local geometry of the motif region itself, and so contribute through shape readout rather than through new direct contacts. At the intermediate scale of tens to hundreds of base pairs, *broader compositional features modulate the nonspecific affinity that governs how a TF samples DNA during 1D diffusion along the helix.* [TODO:check for phrasing in the paper] Sequence correlations and poly(dA:dT) tract content shape this nonspecific affinity landscape (Afek a Lukatsky, 2013; Sela a Lukatsky, 2011), so the sequence environment a scanning TF traverses is not energetically flat. Berg et al. (1981) established that TFs locate their targets faster than three-dimensional diffusion would allow by alternating 3D excursions with 1D sliding along DNA, and Slutsky a Mirny (2004) formalised the *kinetic trade-off between fast scanning and stable specific binding.* A target search of this kind is in principle sensitive to sequence-encoded biases over distances much longer than the binding site itself.

Faltejsková et al. (2026) recently characterised such biases at kilobase scale. In a systematic analysis of in vivo binding sites for ten TFs from ENCODE ChIP-seq, they identified a statistically significant enrichment of GC content spanning approximately 1–2 kb around the binding site for nearly all TFs studied, with magnitudes on the order of ten percentage points [numbers?] and largely symmetric profiles around the peak centre. For a subset of TFs — notably MYC — they additionally observed directional dinucleotide asymmetries: upstream-enriched AA, CA and AC dinucleotides mirrored by downstream-enriched TT, TG and GT, producing a sequence gradient pointing toward the binding site. From these observations the authors propose a statistical funnel model in which broad compositional gradients create a low-free-energy landscape biasing 1D-scanning TFs toward their targets; once near the binding site, short-range readout completes the binding event. A theoretical feature of this proposal is that the gradients extend well beyond the *Debye screening length* over which direct electrostatic protein–DNA interactions decay, so the signal cannot be read out by long-range electrostatic contact and must instead modulate the scanning process itself. This study is *purely descriptive* — the authors propose the funnel as a hypothesis consistent with their observations rather than as a validated mechanism — and they do not test whether any model trained on raw sequence implicitly encodes the gradients they describe. That question motivates the rest of this thesis.

1.1.3 Experimental mapping of binding sites

Throughout this thesis, "binding site"refers to a position identified by ChIP-seq, the standard high-throughput assay for in vivo TF occupancy. The procedure consists of cross-linking protein–DNA contacts in living cells (typically with formaldehyde), fragmenting the chromatin, immunoprecipitating fragments bound by the TF of interest using a specific antibody, and sequencing the recovered DNA. Reads are aligned to the reference genome, where they accumulate as enrichment peaks at bound regions relative to a control library of input chromatin or non-specific IgG immunoprecipitate. Peak calling — the identification of statistically enriched regions against the local background — is conventionally performed with MACS2 (Zhang et al., 2008), and the output is a narrowPeak BED file containing the genomic coordinates of each peak, a significance score, and the offset of the position of maximum read pile-up within the peak (the summit).

Single-replicate peak calls contain false positives. The Irreproducible Discovery Rate framework (Li et al., 2011) addresses this by comparing peak rankings between biological replicates and retaining only peaks that are reproducibly called above a defined irreproducibility threshold. ENCODE provides IDR-thresholded peak files as the standard high-confidence output of its uniform processing pipeline, and these are the files used throughout this thesis as well as by Faltejsková et al. (2026).

A ChIP-seq peak is a region of read enrichment, typically a few hundred base pairs wide, rather than a single binding event. The actual point of TF–DNA contact is somewhere within the peak — most defensibly the summit, more conventionally the peak midpoint. The choice between the two is methodological rather than substantive; *this thesis follows Faltejsková et al. in using the midpoint as the binding-site proxy, and the consequences of the choice are revisited in [odkazMetody?]*. Beyond this positional uncertainty, ChIP-seq has further well-known limitations: antibody quality varies between targets, cross-linking can capture transient or indirect contacts, fragment-length resolution caps the spatial precision at a few hundred base pairs, and any peak set is specific to the cell line and condition in which it was generated.

ENCODE is the data source for the analyses in this thesis. The project provides standardised processing, IDR-thresholded peak files, audit metadata for problematic experiments, and broad coverage of TFs across human cell lines, which together make it the natural reference resource for systematic TF studies — and the reason both Faltejsková et al. and this thesis use it.

1.2 Computational methods for motif analysis

1.2.1 PWMs

[TODO: logo sekvence, vizuál pwm?]

The standard representation of a TF binding motif is the position weight matrix (PWM), constructed from an aligned set of known binding sequences. Counts of each base at each position give the position frequency matrix; dividing by the number of sequences gives the position probability matrix $f_{b,i}$. The PWM entries are the log-odds of these probabilities against a background distribution, $w_{b,i} = \log_2 \left(f_{b,i} / p_b \right)$, where p_b is typically 0.25 for each base or the genome-wide composition. The conservation at each position is summarised by the information content $I_i = 2 - H_i$, where $H_i = -\sum_b f_{b,i} \log_2 f_{b,i}$ is the Shannon entropy of the base distribution at position i. Sequence logos visualise the PWM by stacking letters at each position with heights proportional to $f_{b,i} \cdot I_i$, so that conserved positions appear as tall, near-pure stacks and degenerate positions as short, mixed ones. To identify candidate binding sites in a genome, the PWM is slid across the sequence and each L-bp window is scored by summing the matrix entries corresponding to its bases: $S = \sum_{i=1}^{L} w_{b_i,i}$. Standard scanning tools such as FIMO (Grant et al., 2011) convert this score into a p-value against the background distribution, and report windows below a chosen significance threshold as hits. The additive form of the scoring rule encodes a strong assumption: that each position contributes to binding affinity independently of the others. This approximation is known to be violated for many TFs — for example, where the identity of one position constrains which bases are tolerated at a neighbouring position **[TODO: citaci]** — and motivates the extensions discussed below.

Curated PWM models for known TFs are derived computationally from sets of enriched sequences, typically obtained from in vitro selection assays such as HT-SELEX or from high-confidence ChIP-seq peaks. The two most widely used discovery tools are MEME (Bailey et al., 2015), which identifies over-represented motifs by expectation maximisation, and HOMER (Heinz et al., 2010), which is optimised for ChIP-seq and identifies motifs by differential enrichment against a matched background. Both produce PWMs as their primary output. Curated databases — JASPAR (Ovek Baydar et al.,

2026) for general-purpose use and HOCOMOCO (Vorontsov et al., 2024) for human and mouse TFs — collect such models across studies and provide reference resources for downstream applications; this thesis uses HOCOMOCO v12, matching the choice of (Faltejsková et al., 2026).

Several approaches relax the positional independence assumption of PWMs. Dinucleotide weight matrices extend the framework minimally, by scoring adjacent pairs of bases rather than single positions, and so capturing nearest-neighbour dependencies (Siebert a Söding, 2016); hidden Markov models go further, allowing variable-length and internally structured motifs through state transitions (Eddy, 2004) **[TODO: terrible source and i dont have access, fix!]**. Other extensions — Bayesian networks for arbitrary positional dependencies, and k-mer-based methods that learn from short subsequence frequencies directly — explore the same conceptual space. Benchmarking on in vitro binding data has shown that mononucleotide PWMs trained by the best methods perform comparably to more complex models for the majority of TFs, with measurable gains concentrated in a minority of cases (Weirauch et al., 2013). The fundamental constraint these extensions share is locality: regardless of how the independence assumption is relaxed, the model remains bounded to the motif region itself.

The PWM framework and its extensions are bounded by the binding footprint by design. The kilobase-scale gradients described in **[TODO: inside reference, co s nimi?]** §1.1.4, however, lie well outside that footprint, and have so far been characterised only with hand-engineered compositional features rather than learned directly from sequence. DNA language models — foundation models trained on raw genomic sequence over long contexts — operate at a scale where such gradients fall within their input window, and so could in principle encode them as part of their learned representation. The question this thesis addresses is whether a model of this kind, trained on raw sequence with no compositional features supplied, encodes the kilobase-scale signal that Faltejsková et al. (2026) characterised by other means.

1.3 Foundation models for genomics

Foundation models are large neural networks trained on massive unlabelled datasets using self-supervised objectives, yielding general-purpose representations that downstream tasks can draw on (Bommasani et al., 2022). The two principal objectives are masked language modelling, in which the model recovers hidden tokens from their context, and autoregressive next-token prediction, in which the model predicts each token in sequence given everything to its left. This contrasts with supervised learning, which requires task-specific labels and therefore scales with the cost of annotation rather than with the availability of raw data. In genomics, this approach has produced models trained on raw nucleotide sequence at scale.

1.3.1 the architectures i guess

Genomic foundation models fall into two architectural families that have developed in parallel: those built on the transformer, and those built on alternatives to attention. The first genomic foundation models adapted BERT to nucleotide input: DNABERT (Ji et al.,

2021) with masked language modelling on k-merised human sequence, and Nucleotide Transformer (Dalla-Torre et al., 2025) scaling this across multiple species and parameter counts. Both were limited in the length of input sequence they could process at once. In contrast, HyenaDNA (Nguyen et al., 2023) replaced attention with the Hyena operator, reaching context lengths in the hundreds of thousands of base pairs. Across both families, the design priority has been to extend the context window.

1.3.2 Evo2 and how they used it

Evo 2 (Brixi et al., 2026) is a family of genomic foundation models developed by the Arc Institute, built on the StripedHyena 2 architecture. Attention scales quadratically in sequence length, which limits transformer-only models; StripedHyena 2 addresses this by combining attention with gated convolutional operators that scale linearly while preserving long-range dependencies. Evo 2's context window extends to 1 Mb at single-nucleotide resolution, comfortably hosting the kilobase-scale gradients of §1.1.4 within a single input. Evo 2 was pretrained autoregressively on OpenGenome2, a dataset of 8.8 trillion DNA tokens drawn from bacteria, archaea, eukarya, and bacteriophage. Pretraining proceeded in two phases: a short-context phase on roughly 8 kb windows enriched for genic regions, followed by a long-context phase extending to 1 Mb on whole-genome samples. Three capabilities follow: sequence scoring, which assigns a likelihood to an input under the model; generation, which samples new sequences; and embedding extraction, which retrieves representations from intermediate layers.

[TODO: simplified StripedHyena 2 diagram, or the one from the paper]

[TODO: what they did with TFs, what am i doing?]

1.4 NGS data

Chapter 2

Aims of thesis

1.

Chapter 3

Methods $\int f(x)\,\mathrm{d}x$ v názvu

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odkaz na rovnici s „tagem“ (rovnice) – pokud nechcete vzorci přidělit číslo, ale nějaký vlastní symbol, používejte „hvězdičkovaná“ prostředí, tj. např. `equation*`

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Chapter 4

Druhá kapitola s matematikou $\int f(x)\,\mathrm{d}x$ v názvu

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$$\int f(x)\,\mathrm{d}x \tag{rovnice}$$

odkaz na rovnici s „tagem“ (rovnice) – pokud nechcete vzorci přidělit číslo, ale nějaký vlastní symbol, používejte „hvězdičkovaná“ prostředí, tj. např. `equation*`

$$\iint f(x)\,\mathrm{d}x \tag{4.1}$$

odkaz na druhy vzorec (4.1)

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4.1 Podkapitola

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Závěr

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Appendix

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